Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Diode characteristics, Shockley's equation

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Diode characteristics, Shockley's equation

1 Diode characteristics, Shockley's equation

Study focus: Diode characteristics, Shockley's equation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Professor Ananda, Department of Electronics and Communication Engineering.

Key details

- Today, we are going to see the semiconductor diode characteristics and parameters and with some numericals.
- As we look at the forward bias here, initially for the applied voltage, the diode was off, and at particular voltage of Vf, the current start increasing.
- Then, we also come to another parameter called forward current rating.
- The forward current rating, which is I.
- It's the maximum current at which the diode starts conducting.
- It depends on the voltage at which the diode is on.
- So the current can vary based on the device, because it depends on the fabrication of the device.
- Then the maximum power expression P is equal to V into I.
- The first is breakdown voltage.
- Until there is a breakdown, there is a leakage current flowing, which is a reverse saturation current, which is less than the reverse saturation current, which is called as a leakage current.
- We need to remember this.
- Then, the reverse saturation current is this region, where the diode start conducting even with the reverse bias.
- That happens at a reverse bias voltage.
- It could vary based on the device.
- It could be five volts reverse saturation current.

- That depends on the fabrication again.
- Then we also need to calculate what the peak inverse voltage, the voltage at which the device starts conducting, which is a peak inverse voltage rating.
- And the diode representation symbol is this, where it looks like an arrow lying here.
- The arrow indicate the conventional current flow from P region to N region.
- Next, we also learn here the diode current expression or the Shockley's current equation.
- When Vd is positive, we represent Id is equal to minus Is.
- When Vd is negative, we represent Id is equal to minus Is.
- When Vd is positive, we represent Id is approximately equal to Is into E to the power of qVd by NmkBT, where Id and Vd are the diode current and voltage respectively, where q is the charge of the electron, where eta is the ideality factor, where eta N is equal to one for indirect semiconductor like silicon, germanium, et cetera.
- For a direct semiconductor, N is equal to two, where gallium arsenide, if you use gallium arsenide and indium phosphate.
- Where k is the Boltzmann constant, T is the temperature in Kelvin.
- We are going to use these k and the q values.
- Where the kT by q is also called as a thermal voltage, where kT by q is equal to 25.9 millivolts.
- At a temperature 27 degrees centigrade, a common temperature for a component in an enclosed operating system, determine the thermal voltage.
- As we know the thermal voltage, how to calculate?
- VT is equal to kT by q, where we need to calculate the T here.

Example or application from the lecture

- The Shockley's equation is also called as a diode equation, help us to determine the current at the given temperature, knowing the diode voltage and saturation current.
- That is given by Id is equal to Is into qVd by NqT minus one.
- Then, k is the Boltzmann constant given by value.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

Then substitute that here, where we know the k value, 1.38 into 10 to the power of -23, multiply with 300 Kelvin here, divide by 1.6 into 10 to the power of -19 coulombs.

Key details

- Then coming to the numerical two, calculate the thermal voltage when the temperature is 25 degree.
- The previous numerical was at 27 degree, now calculate at 25 degree.
- Similarly, coming to the next numerical.
- Calculate the forward bias current of silicon diode when forward voltage is 0.4 applied and the reverse saturation current is 1.17 and the thermal voltage is 25.2 millivolt.
- We need to calculate Id here.

- Since in this equation, ideality factor is not mentioned, it can be taken as one.
- Coming to numerical three.
- Vd NK T minus one.
- We have to find VT thermal voltage and reverse saturation current.
- Then substitute all the values for Id, for Is.
- For Id, you have to calculate.
- For Is, we need to calculate.
- The summary is the current should be less compared to the forward current.
- The reverse current is pico in 10 to the power of minus 12.
- Coming to the numerical four, calculate the applied voltage Vd if the diode current is six milliamps, thermal voltage is 26 millivolts, ideality is one, reverse saturation current is one.
- This numerical is not similar to the previous what we solved.
- Next, coming to the numerical five.
- Find the change in the voltage if current changes from 0.1 milliamps to 10 milliamps.
- We need to calculate 26 milli.
- We can directly use that value 26 ln 10, and the change, the I2 by I1.
- 10 milli by 0.1, so we get 0.143 volts.
- And finally, we can see the comparison of the diode characteristics of germanium, silicon, and gallium arsenide with respect to the forward cutting voltages.
- And also the reverse saturation current also has been shown.
- The breakdown voltage will vary for type of the device.

Example or application from the lecture

- Vd is equal to, which is kT by q, with a given value, they have already given 25.2, we know that n is equal to one, Vd is equal to 0.4.
- Substitute all the values here and calculate Id, which is given by 9.156 milliamps, and which is going to be solved in a class by the faculty.
- Given a diode current of eight milli and N is equal to one, find the reverse saturation current if the diode voltage is 0.5 and the room temperature is 25 degree centigrade.

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Core explanation · 10:00–14:59

We see that the breakdown voltage for germanium is very high, 100 volts.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Diode characteristics, Shockley's equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...

C Revision and retrieval practice